

Precalc Lesson 1.3 Homework

Answer Key

1. -4

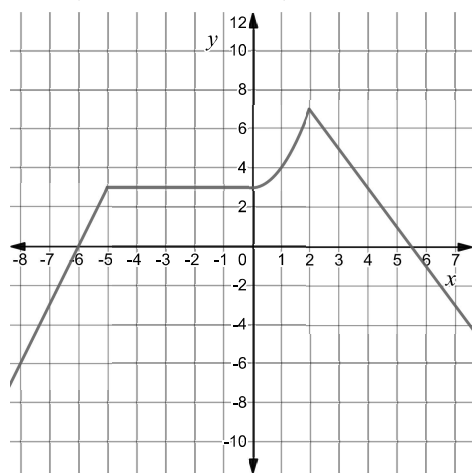
2. $-4.9^\circ \text{ F per minute}$

3. $-\frac{3}{5}$

4. C, A, D, B

5. f is increasing on $[-1.21, 1.4]$ and $[3.56, \infty)$
 f is decreasing on $(-\infty, -1.21]$ and $[1.4, 3.56]$
 f is never constant.

6. Many answers are possible. The graph shows one possible answer.



7. a. $f(x)$ has a relative minimum at $x = -3$.

b. There is no relative maxima.

8. B

9. a. \$55,000,000
b. \$100,000
c. Yes, if they spend \$204,880 dollars on advertising, they will make no profit.
10. a. a is increasing on $[0, 3]$ which means that the rate at which people enter the art museum increases during the first 3 hours it is open.
b. a is constant on $[4, 6]$ which means that people are entering the museum at a constant rate between $t = 4$ and $t = 6$ hours.
c. The graph of a has a maximum value of 180 people per hour. This means that the fastest rate at which people enter the museum is 180 people per hour and this occurs at $t = 3$ hours.
d. $a(t) = 0$ when $t = 8$. This means that no people are entering the museum at $t = 8$ hours, which probably means that the museum is closed.